
Using the R_{BIAS} Pin of Single Positive Supply RF Amplifiers as a Fast Enable/Disable Input
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INTRODUCTION

RF amplifiers designed in the traditional depletion mode gallium arsenide (GaAs) pseudomorphic high electron mobility transistor (pHEMPT) process require a negative gate voltage to set the drain current. Recent semiconductor process enhancements and design innovations have given rise to RF amplifiers that operate from a

single positive supply voltage and are biased by a single-supply referenced resistor. While there is an obvious benefit to eliminating negative supply voltages, the R_{BIAS} function can also be used to turn on and off the device. This article explores these capabilities and presents some of the associated trade-offs.

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REVISION HISTORY

12/2024—Revision 0: Initial Version

TRADITIONAL NEGATIVE GATE VOLTAGE BIASING

Figure 1 shows the HMC8410, a low noise RF amplifier that operates from 10MHz to 10GHz. It requires a drain current, set by the negative gate voltage, to turn on. The application schematic shown in Figure 1 has a positive drain voltage that is applied through an inductor to the RF_{OUT}/V_{DD} pin. A negative gate voltage is applied directly to the RF_{IN}/VGG1 pin via an inductor. The negative gate voltage is adjusted to set the desired drain current. To avoid excessive drain current and burnout, the negative gate bias voltage must be applied before the drain voltage.

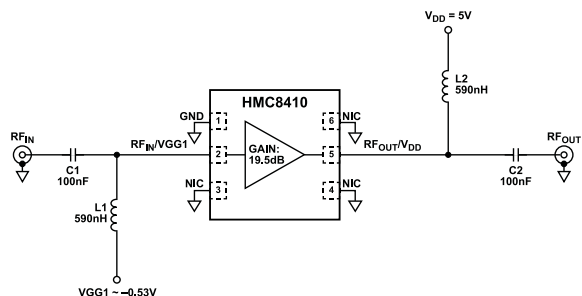


Figure 1. Setting the Drain Current of the HMC8410 Low Noise Amplifier (LNA) Using a Negative Gate Bias Voltage

SINGLE POSITIVE SUPPLY RF AMPLIFIER WITH RESISTOR ADJUSTABLE BIAS CURRENT

Figure 2 shows a single positive supply low noise RF amplifier, the HMC8412, which operates at 400MHz to 11GHz. In this schematic, a resistor is placed between the R_{BIAS} pin and the drain voltage to set the desired drain current. Figure 3 shows a plot of the bias current vs. resistor values at different supply voltages. The typical current flowing through the R_{BIAS} resistor is a few milliamperes.

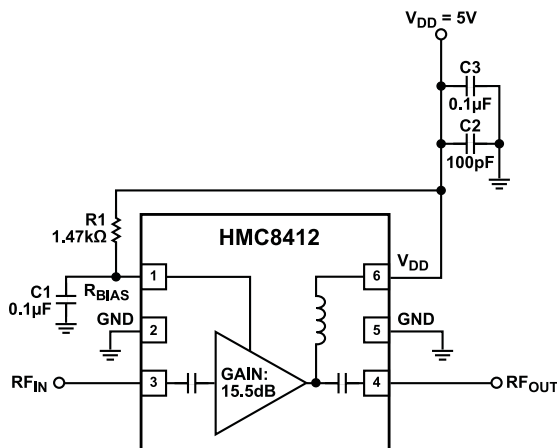


Figure 2. Single Positive Supply Using R_{BIAS} Resistor

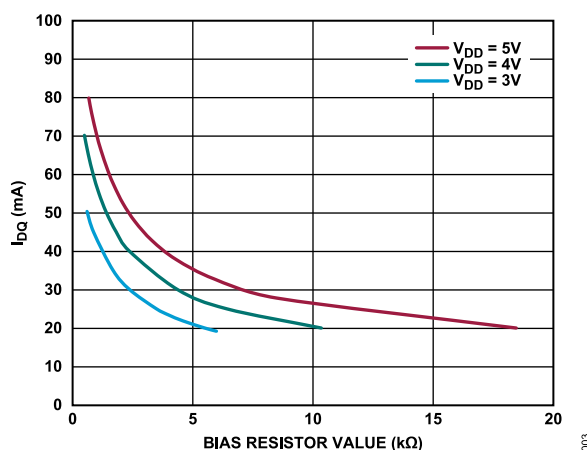


Figure 3. HMC8412 400MHz to 11GHz LNA, Drain Current (IDQ) vs. Bias Resistor (R_{BIAS}) for Various V_{DD} Values (R_{BIAS} Connected to V_{DD})

USING THE R_{BIAS} PIN AS AN ENABLE/DISABLE INPUT

The ability for an RF amplifier to shut down is a desirable feature in power sensitive applications. In Figure 4, the ADG719, a single-pole double throw (SPDT) switch is used to disable and enable the HMC8412, by connecting the R_{BIAS} resistor either to supply ground. Featuring an on and off time of 12ns and 6ns, respectively, the ADG719 switch is not a limiting factor in the overall response time.

If the enable/disable signal is coming from a 50Ω source, it is good practice to terminate at the input, IN, of the switch with a 50Ω resistor. An alternative approach is to use a single-pole single throw (SPST) switch to turn on and off the amplifier by switching the R_{BIAS} pin directly to ground. A disadvantage of this approach is that the current continues to flow through the R_{BIAS} resistor.

Figure 4. Using an SPDT Analog Switch and the R_{BIAS} Resistor to Enable and Disable the HMC8412, 400MHz to 11GHz LNA

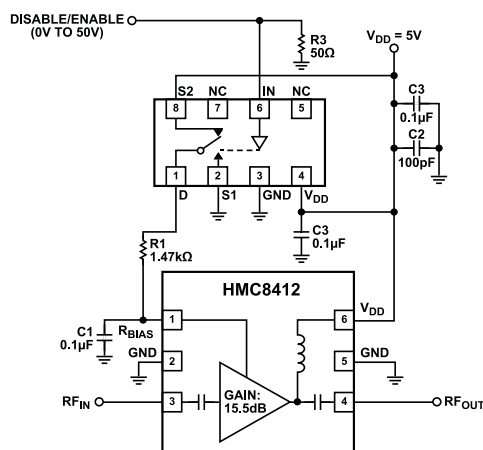


Figure 5 shows the response time of the circuit in Figure 4. The blue trace is the enable signal, and the green trace is the RF_{OUT} envelope. The turn-on response is approximately 200ns, and the turn-off response is approximately 500ns.

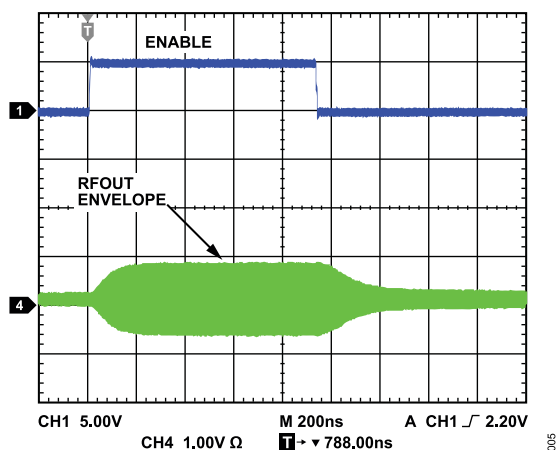


Figure 5. On/Off Response Time of the Circuit in Figure 4 (Blue = R_{BIAS} Enable, Green = RF_{OUT} Envelope)

IMPACT OF MATCHING NETWORK ON ENABLE/DISABLE RESPONSE TIME

In Figure 6, the ADG719 is used to enable and disable the HMC8413, a low noise RF amplifier that operates from 10MHz to 9GHz. This circuit includes a shunt resistor, inductor, and capacitor (RLC) network at the input of the HMC8413. This RLC network adds resistive loss that reduces the gain at low frequencies and stabilizes the amplifier.

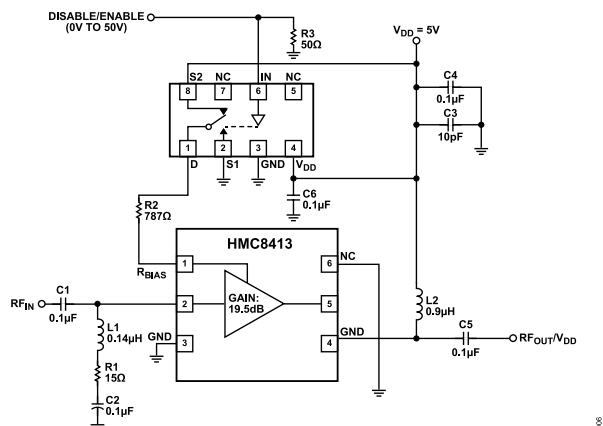


Figure 6. Using an SPDT Analog Switch and the R_{BIAS} Resistor to Enable and Disable the HMC8413, 10MHz to 9GHz LNA

Figure 7 shows the response time for the circuit in Figure 6. The turn-on response is approximately 240 μ s, and the turn-off response is approximately 840 μ s. This response time is significantly slower than the response time seen for the HMC8412 application circuit (Figure 5).

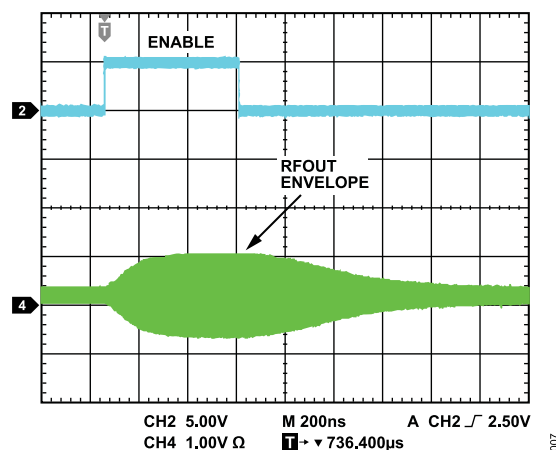


Figure 7. On/Off Response Time of the Circuit in Figure 6 (Teal = R_{BIAS} Enable, Green = RF_{OUT} Envelope)

MODIFIED CIRCUIT WITH ENHANCED RESPONSE TIME

The response time can be improved by eliminating the RLC network as shown in Figure 8. In addition, the AC-coupling capacitors and bias inductor are reduced in size. Modifications to the power supply decoupling components are also made to include a 5 Ω resistor to ensure stability in the absence of the RLC network.

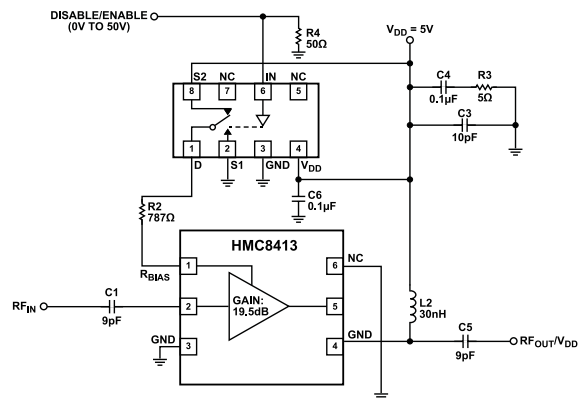


Figure 8. R_{BIAS} Enable/Disable Circuit Modifications to Improve Response Time

Figure 9 shows the response time of the modified circuit. The turn-on response is approximately 200ns, and the turn-off response is approximately 240ns. This response time is much faster than the response time seen for the default HMC8413 application circuit (Figure 7).

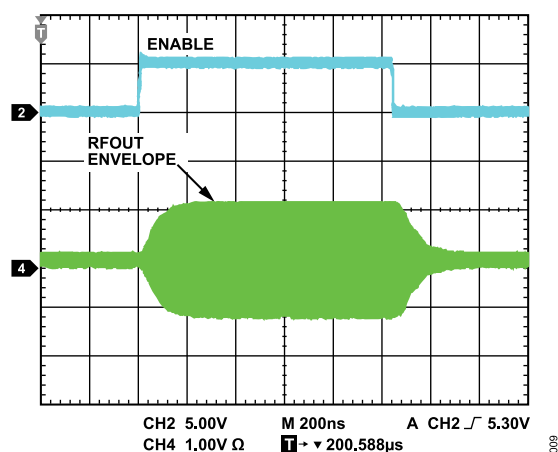


Figure 9. Response Time of Modified R_{BIAS} Enable/Disable Circuit, Optimized for Fast Response Time

S-PARAMETER COMPARISON OF ORIGINAL AND MODIFIED CIRCUITS

Figure 10 shows the S-parameters of the original and modified circuits. The modified circuit has a low-frequency response that rolls off at 200MHz. This is the result of the smaller AC coupling capacitors and bias inductor. The gain for both circuits is similar at frequencies above 600MHz.

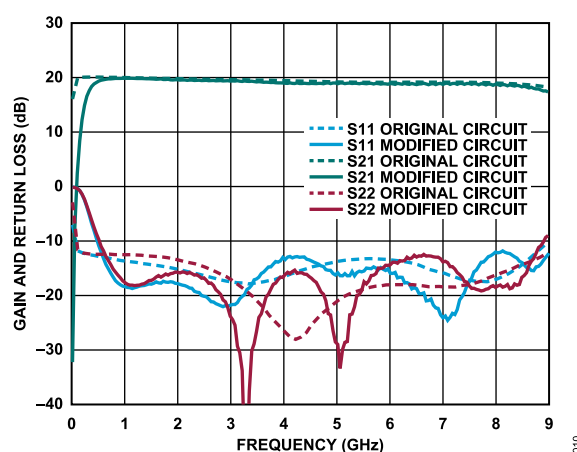


Figure 10. S-Parameters of Original and Modified Circuit

CONCLUSION

The advent of single positive supply RF amplifiers with adjustable biasing is a significant advantage because it removes the need for negative gate supply voltages and the associated on and off sequencing. While all newer Analog Devices, Inc., LNAs have this functionality, going forward, higher power driver amplifiers and power amplifiers come available with single positive supplies. The R_{BIAS} function can be used as a convenient shutdown feature, but some circuit modifications may be required if a fast turn-on and -off response is required.